

宁波大学
NINGBO UNIVERSITY

实 验 报 告

实验 12 数据备份与恢复

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实验项目名称	实验 12 数据备份与恢复

一、 实验目的

本实验的目的是了解并掌握 **OpenGauss** 数据库的备份与恢复机制，深入理解物理和逻辑备份的区别及应用场景，能够通过实际操作完成数据库的备份与恢复。同时，通过学习 **MOT（Memory-Optimized Tables）**的基本应用，掌握将磁盘表转换为内存优化表的操作方法，提高对数据库性能优化和数据管理的实践能力。

二、 实验内容及过程、结果

1.物理备份与恢复

（1）物理热备份所有数据库并恢复数据库

① 导出所有数据库。

首先创建 **data** 文件夹，并进入数据库

```
[root@db1 ~]# cd /home
[root@db1 home]# mkdir data
[root@db1 home]# chmod 777 data
[root@db1 home]# su - omm
Last login: Fri Dec  6 19:46:18 CST 2024 on tty1


Welcome to 4.19.90-2003.4.0.0036.oe1.x86_64

System information as of time:  2024年 12月 06日 星期五 19:51:16 CST

System load:    0.19
Processes:      108
Memory used:    10.6%
Swap used:      0.5%
Usage On:       41%
IP address:     10.0.2.15
Users online:   3
```

```
[omm@db1 ~]$ gs_om -t start
Starting cluster.
=========================================
=========================================
Successfully started.
[omm@db1 ~]$ gs_basebackup -D /home/data -h 127.0.0.1 -p 26000
INFO:  The starting position of the xlog copy of the full build is: 0/
3000028. The slot minimum LSN is: 0/0.
begin build tablespace list
finish build tablespace list
begin get xlog by xlogstream

 check identify system success

 send START_REPLICATION 0/3000000 success

 keepalive message is received

 keepalive message is received

 keepalive message is received
gs_basebackup: base backup  successfully
[omm@db1 ~]$
```

② 恢复数据库。

```
[omm@db1 ~]$ cp -r /home/data  /gaussdb/data/db1
[omm@db1 ~]$
```

2.逻辑备份与恢复

(1) 导出 **opengauss** 所有数据库并恢复某个数据库

①导出 **opengauss** 所有数据库**:**

```
[root@db1 ~]# cd /home
[root@db1 home]# mkdir backup
[root@db1 home]# chmod 777 backup
[root@db1 home]# su - omm
Last login: Fri Dec  6 20:11:27 CST 2024 on pts/0


Welcome to 4.19.90-2003.4.0.0036.oe1.x86_64

System information as of time:  2024年 12月 06日 星期五 20:16:00 CST

System load:    0.58
Processes:      109
Memory used:    29.2%
Swap used:      1.1%
Usage On:       45%
IP address:     10.0.2.15
Users online:   3
```

```
[omm@db1 ~]$ gs_dumpall -f /home/backup/backup.sql -p 26000
gs_dump[port='26000'][dbname='db_tpcc'][2024-12-10 22:33:09]: The tota
l objects number is 386.
gs_dump[port='26000'][dbname='db_tpcc'][2024-12-10 22:33:09]: [100.00%
] 386 objects have been dumped.
gs_dump[port='26000'][dbname='db_tpcc'][2024-12-10 22:33:09]: dump dat
abase dbname='db_tpcc' successfully
gs_dump[port='26000'][dbname='db_tpcc'][2024-12-10 22:33:09]: total ti
me: 372  ms
gs_dump[port='26000'][dbname='db_uni'][2024-12-10 22:33:09]: The total
 objects number is 487.
gs_dump[port='26000'][dbname='db_uni'][2024-12-10 22:33:09]: [100.00%]
 487 objects have been dumped.
gs_dump[port='26000'][dbname='db_uni'][2024-12-10 22:33:09]: dump data
base dbname='db_uni' successfully
gs_dump[port='26000'][dbname='db_uni'][2024-12-10 22:33:09]: total tim
e: 650  ms
gs_dump[port='26000'][dbname='postgres'][2024-12-10 22:33:10]: The tot
al objects number is 408.
gs_dump[port='26000'][dbname='postgres'][2024-12-10 22:33:10]: [100.00
%] 408 objects have been dumped.
gs_dump[port='26000'][dbname='postgres'][2024-12-10 22:33:10]: dump da
tabase dbname='postgres' successfully
gs_dump[port='26000'][dbname='postgres'][2024-12-10 22:33:10]: total t
ime: 310  ms
gs_dumpall[port='26000'][2024-12-10 22:33:10]: dumpall operation succe
ssful
gs_dumpall[port='26000'][2024-12-10 22:33:10]: total time: 1614  ms
[omm@db1 ~]$
```

②恢复 **postgres** 数据库: **gsql -p 26000 -f /home/backup/backup.sql postgres**

```
[omm@db1 ~]$ gsql -p 26000 -f /home/backup/backup.sql postgres
SET
SET
gsql:/home/backup/backup.sql:12: ERROR:  role "CeduManagerRole" alread
y exists
ALTER ROLE
gsql:/home/backup/backup.sql:14: ERROR:  role "CeduSA" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:16: ERROR:  role "CeduSJ" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:18: ERROR:  role "StuG" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:20: ERROR:  role "StuL" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:22: ERROR:  role "StudentRole" already ex
ists
ALTER ROLE
gsql:/home/backup/backup.sql:24: ERROR:  role "TeaW" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:26: ERROR:  role "TeaY" already exists
ALTER ROLE
gsql:/home/backup/backup.sql:28: ERROR:  role "TeacherRole" already ex
ists
ALTER ROLE
gsql:/home/backup/backup.sql:30: ERROR:  role "UeduManagerRole" alread
y exists
ALTER ROLE
```

(2) 备份和还原数据库

①将 **student_mis** 数据库备份到主机 **127.0.0.1 /home/data**。在 **127.0.0.1** 主机执行以下命令：

```
[root@db1 ~]# cd /home
[root@db1 home]# mkdir data
[root@db1 home]# chmod 777 data
[root@db1 home]# su - omm
Last login: Fri Dec  6 19:46:18 CST 2024 on tty1


Welcome to 4.19.90-2003.4.0.0036.oe1.x86_64

System information as of time:  2024年 12月 06日 星期五 19:51:16 CST

System load:    0.19
Processes:      108
Memory used:    10.6%
Swap used:      0.5%
Usage On:       41%
IP address:     10.0.2.15
Users online:   3
```

```
[omm@db1 ~]$ gs_dump -U nbuuser -W Rxb040611@ -h 127.0.0.1 -p 26000 st
udent_mis -f /home/data/student.sql -F c
gs_dump[port='26000'][student_mis][2024-12-10 23:43:52]: The total obj
ects number is 386.
gs_dump[port='26000'][student_mis][2024-12-10 23:43:52]: [100.00%] 386
 objects have been dumped.
gs_dump[port='26000'][student_mis][2024-12-10 23:43:52]: dump database
 student_mis successfully
gs_dump[port='26000'][student_mis][2024-12-10 23:43:52]: total time: 4
77  ms
[omm@db1 ~]$
```

② 将内容还原到 **student_mis** 数据库。在 **127.0.0.1** 主机执行以下命令：

```
[omm@db1 ~]$ gs_restore -U nbuuser -W Rxb040611@ -h 127.0.0.1 -p 26000
 -d student_mis /home/data/student.sql
start restore operation ...
Finish reading 6 SQL statements!
end restore operation ...
restore operation successful
total time: 84  ms
[omm@db1 ~]$
```

(3)导出和导入数据表 **public.reports**

①导出 **public.reports:**

```
[root@db1 ~]# cd /home
[root@db1 home]# mkdir table
[root@db1 home]# chmod 777 table
[root@db1 home]#
```

```
[omm@db1 ~]$ gs_dump -W Rxb040611@ -U nbuuser -f /home/table/reports.d
mp -p 26000 student_mis -t public.reports  -F c
gs_dump[port='26000'][student_mis][2024-12-11 00:08:16]: The total obj
ects number is 382.
gs_dump[port='26000'][student_mis][2024-12-11 00:08:16]: [100.00%] 382
 objects have been dumped.
gs_dump[port='26000'][student_mis][2024-12-11 00:08:16]: dump database
 student_mis successfully
gs_dump[port='26000'][student_mis][2024-12-11 00:08:16]: total time: 1
97  ms
[omm@db1 ~]$
```

②导入 **public.reports:**

```
[omm@db1 ~]$ gs_restore /home/table/reports.dmp -p 26000 -d student_mi
s
start restore operation ...
Error while PROCESSING TOC:
Error from TOC entry 9; 1259 41255 TABLE reports nbuuser
could not execute query: ERROR:  relation "reports" already exists
    Command was: CREATE TABLE reports (
    id integer DEFAULT nextval('reports_id_seq'::regclass) NOT NULL,
    sno character varying(10) NO...
Error from TOC entry 3643; 0 41255 TABLE DATA reports nbuuser
COPY failed for table "reports": ERROR:  duplicate key value violates
unique constraint "reports_pkey"
DETAIL:  Key (id)=(1) already exists.
CONTEXT:  COPY reports, line 4: ""
table reports complete data imported !
Error from TOC entry 3642; 2606 41260 CONSTRAINT reports_pkey nbuuser
could not execute query: ERROR:  multiple primary keys for table "repo
rts" are not allowed
    Command was: ALTER TABLE reports
    ADD CONSTRAINT reports_pkey PRIMARY KEY (id);



Finish reading 5 SQL statements!
end restore operation ...
WARNING: errors ignored on restore: 3
restore operation successful
total time: 19  ms
[omm@db1 ~]$
```

(4) 备份和还原数据库主机 **db1**

①备份：**gs_backup - t backup --backup-dir=/home/data -h db1 – all**

```
[omm@db1 ~]$ gs_backup -t backup --backup-dir=/home/data -h db1 --all
Parsing configuration files.
Successfully parsed the configuration file.
Performing remote backup.
[FAILURE] db1:
Parsing the configuration file.
Obtaining local installation path for backup.
Successfully parsed the configuration file.
Checking backup directory.
Successfully checked backup directory.
Backing up files.
Backing up binary files.
Successfully backed up binary files.
Backing up parameter files.
[GAUSS-51400] : Failed to execute the command: pscp -x '' -H db1 /opt/
huawei/wisequery/omm_mppdb/backupTemp_829441/parameter_db1.tar /opt/hu
awei/wisequery/omm_mppdb/backupTemp_829441 . Error:
[1] 01:23:48 [FAILURE] db1 Exited with error code 1
[omm@db1 ~]$
```

②还原：gs_backup　- t restore　--backup-dir=/home/data -h db1 –all

```
[omm@db1 ~]$ gs_backup -t restore --backup-dir=/home/data -h db1 --all
Parsing configuration files.
Successfully parsed the configuration file.
Performing remote restoration.
[FAILURE] db1:
[GAUSS-50201] : The Binary files does not exist.
Executing the local restoration
Parsing the configuration file.
Getting local installation path for restoration.
Successfully parsed the configuration file.
Checking restored directory.
Successfully checked restored directory.
Restoring files.
Restoring binary files.
[omm@db1 ~]$
```

3.使用 MOT---将磁盘表转换为 MOT

①创建 **/home/table** 目录

```
[root@db1 ~]# cd /home
[root@db1 home]# mkdir table
[root@db1 home]# chmod 777 table
[root@db1 home]#
```

②修改 **postgresql.conf　enable_incremental_checkpoint = off**

```
[root@db1 /]# cd /gaussdb/data/db1
[root@db1 db1]# vi postgresql.conf
```

```
enable_incremental_checkpoint 1 of 1
# - Checkpoints -

checkpoint_segments = 64                # in logfile segments, min 1,
16MB each
#checkpoint_timeout = 15min             # range 30s-1h
#checkpoint_completion_target = 0.5     # checkpoint target duration,
0.0 - 1.0
#checkpoint_warning = 5min              # 0 disables
#checkpoint_wait_timeout = 60s  # maximum time wait checkpointer to st
art

enable_incremental_checkpoint = off     # enable incremental checkpoin
t
incremental_checkpoint_timeout = 60s    # range 1s-1h
#pagewriter_sleep = 100ms               # dirty page writer sleep time
, 0ms - 1h
```

③ 将数据库 **postgres** 中 **reports** 数据表迁移到 **MOT** 中

```
[root@db1 home]# su - omm
Last login: Wed Dec 11 01:22:58 CST 2024 on pts/0


Welcome to 4.19.90-2003.4.0.0036.oe1.x86_64

System information as of time:  2024年 12月 11日 星期三 02:33:16 CST

System load:    0.34
Processes:      111
Memory used:    23.2%
Swap used:      0.7%
Usage On:       51%
IP address:     10.0.2.15
Users online:   3



[omm@db1 ~]$ gsql -h 127.0.0.1 -p 26000 -U nbuuser -d postgres
Password for user nbuuser:
gsql ((openGauss 1.1.0 build 392c0438) compiled at 2020-12-31 20:08:21
 commit 0 last mr  )
SSL connection (cipher: DHE-RSA-AES128-GCM-SHA256, bits: 128)
Type "help" for help.

postgres=> GRANT USAGE ON FOREIGN SERVER mot_server TO nbuuser;
GRANT
postgres=>
```

```
[omm@db1 ~]$ gs_dump -W Rxb040611@ -U nbuuser -f /home/table/reports.dmp -p  2
6000 student_mis -t public.reports -F c
gs_dump[port='26000'][student_mis][2024-12-11 02:53:46]: The total objects num
ber is 382.
gs_dump[port='26000'][student_mis][2024-12-11 02:53:46]: [100.00%] 382 objects
 have been dumped.
gs_dump[port='26000'][student_mis][2024-12-11 02:53:46]: dump database student
_mis successfully
gs_dump[port='26000'][student_mis][2024-12-11 02:53:46]: total time: 188  ms
[omm@db1 ~]$
```

```
postgres=> alter table reportstable rename to reports;
ALTER TABLE
```

```
postgres=> create FOREIGN table reports(sno character varying (10) NOT NULL,cn
o character varying(6) NOT NULL,grade integer);
CREATE FOREIGN TABLE
```

```
[omm@db1 ~]$ gs_restore  /home/table/reports.dmp -p 26000 -d postgres
start restore operation ...
Error while PROCESSING TOC:
Error from TOC entry 9; 1259 41255 TABLE reports nbuuser
could not execute query: ERROR:  relation "reports_id_seq" does not exist
    Command was: CREATE TABLE reports (
    id integer DEFAULT nextval('reports_id_seq'::regclass) NOT NULL,
    sno character varying(10) NO...
could not execute query: ERROR:  relation "public.reports" does not exist
    Command was: ALTER TABLE public.reports OWNER TO nbuuser;


Error from TOC entry 3643; 0 41255 TABLE DATA reports nbuuser
could not execute query: ERROR:  relation "reports" does not exist
    Command was: COPY reports (id, sno, tno, cno, score) FROM stdin;

Error from TOC entry 3642; 2606 41260 CONSTRAINT reports_pkey nbuuser
could not execute query: ERROR:  relation "reports" does not exist
    Command was: ALTER TABLE reports
    ADD CONSTRAINT reports_pkey PRIMARY KEY (id);



Finish reading 5 SQL statements!
end restore operation ...
WARNING: errors ignored on restore: 4
restore operation successful
total time: 68  ms
[omm@db1 ~]$
```

④删除名为 **reports** 的 **MOT**：

```
postgres=> drop FOREIGN table reports;
DROP FOREIGN TABLE
```

三、　实验中遇到的问题及分析讨论

```
[opengauss@530bb0d84ad2 ~]$ gs_basebackup -D /home/data -h 192.168.56.1 -p 5432
gs_basebackup: could not connect to server: could not connect to server: Operation now in progress
        Is the server running on host "192.168.56.1" and accepting
        TCP/IP connections on port 5433?

[opengauss@530bb0d84ad2 ~]$
```

```
[opengauss@530bb0d84ad2 data]$ gs_basebackup -D /home/data -h 192.168.56.1 -p 5431
gs_basebackup: could not connect to server: FATAL:  Forbid remote connection with trust method!
```

Q1:使用 **docker** 方式产生的私网 **ip** 和端口号无法正确进行连接

A1：使用虚拟机下载 **opengauss** 进行实验

Q2：先前已经修改 **postgresql.conf** 使得 **enable_incremental_checkpoint = off,** 打开数据库执行以下命令失败

create FOREIGN table reports (

sno character varying (10) NOT NULL,

cno character varying(6) NOT NULL,

grade integer

);

```
postgres=> create FOREIGN table reports (sno character varying (10) NOT NULL,c
no character varying(6) NOT NULL,grade integer);
ERROR:  Cannot create MOT tables while incremental checkpoint is enabled.
postgres=> alter system set enable_incremental_checkpoint='off';
NOTICE:  please restart the database for the POSTMASTER level parameter to tak
e effect.
```

A2：使用 **restart** 命令重启数据库，在执行以上命令。

```
[omm@db1 ~]$ gs_om -t restart
Stopping cluster.
=========================================
Successfully stopped cluster.
=========================================
End stop cluster.
Starting cluster.
=========================================
=========================================
Successfully started.
```

```
postgres=> create FOREIGN table reports(sno character varying (10) NOT NULL,cn
o character varying(6) NOT NULL,grade integer);
CREATE FOREIGN TABLE
```

四、 实验结论

通过本次实验，我全面学习并掌握了 **OpenGauss** 数据库中备份与恢复的关键技术，包括物理备份与逻辑备份的实际操作方法及其适用场景的区别。在实验过程中，通过实际操作完成了数据库的备份与恢复任务，例如备份所有数据库、恢复指定数据库以及导入导出特定数据表等，进一步加深了对数据库数据安全管理的理解。此外，还学习了将磁盘表转换为内存优化表（**MOT**）的方法，感受到这一技术对数据库性能优化的重要性。在实验过程中，针对 **Docker** 环境的网络连接问题以及配置文件生效问题，通过分析原因并采取适当措施顺利解决。这次实验让我对数据库备份与恢复的原理和实践操作有了更深刻的认识，同时也增强了我解决实际问题的能力，为日后实际项目中的数据库管理提供了宝贵经验。

五、　　教师批阅意见：   成绩评定：   教师签字： 年　　月　　日
六、　　备注：